

SAI HARSHA VEMULA

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📁 [Portfolio](#)

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Software Engineer with 3+ years of experience delivering robust backend services and automating cloud infrastructure using Java, Python, AWS, and Kubernetes. Skilled in building microservices, streamlining CI/CD pipelines, and enabling high-velocity deployments through DevOps best practices.

Education

University of Missouri-Kansas City

Master in Computer Science

August 2023 – December 2024

Kansas City, Missouri

Vellore Institute of Technology

M Tech Integrated Software Engineering

June 2016 – June 2021

Vellore, Tamil Nadu, India

Technical Skills

Languages: Java, Python, Shell Scripting, SQL, CSS, HTML, JavaScript

Database Technologies: AWS RDS, MySQL, PostgreSQL, Elasticsearch, Redis

DevOps Tools & CI/CD: GitLab, Jenkins, SonarQube, Argo CD, Velero

Containerization and Orchestration: Docker, Kubernetes

AWS Cloud Services: EKS, S3, EC2, IAM, RDS, RedShift, VPC, Cloudwatch, SNS, ECS, Route 53, WAF

Azure Cloud Services: Azure Virtual Machines, Azure App Service, AKS, Azure Functions, Azure AD

Security & Secrets Management: HashiCorp Vault, Trivy

Protocols: TCP, UDP, IP, HTTP, HTTPS, SSH, DNS, SFTP

Monitoring & Logging: Prometheus, Grafana, Datadog, ELK Stack (Elasticsearch, Logstash, Kibana)

Infrastructure as Code (IaC): Terraform, AWS CloudFormation, Ansible

Build Tools: Maven, Gradle

Web Servers: Tomcat, Nginx, Apache HTTP Server

Operating Systems: Linux (Ubuntu, CentOS), Windows

Experience

Vitrana, Inc.

Senior Software Engineer - DevOps

July 2021 – June 2023

Bangalore, Karnataka, India

- Directed the development and refinement of Kubernetes manifests, streamlining automated deployment, scaling, and management of containerized applications, boosting deployment speed by **20%** and cutting system downtime by **30%**.
- Designed and managed multiple AWS VPCs, configuring subnets, CIDR allocations, and Network ACLs (NACLs) to enhance network segmentation, security, and performance across cloud environments.
- Deployed robust monitoring and alerting systems using Prometheus and Grafana, slashing incident response times by **40%** and elevating service uptime to **80%**.
- Utilized Docker Compose for containerizing microservices, guaranteeing consistent and repeatable application deployment across all environments, reducing deployment issues by **35%**.
- Applied Infrastructure as Code (IaC) with Terraform, automating cloud infrastructure provisioning, which halved setup time and eradicated configuration drift.
- Fortified system reliability by implementing automated failover strategies and disaster recovery processes, attaining max service uptime.
- Designed and implemented an efficient cache management strategy using Redis and AWS ElastiCache, improving application performance and reducing database load by optimizing data retrieval.
- Established security best practices by integrating Trivy, JUnit, and SonarQube into CI/CD workflows, enhancing code quality and mitigating security vulnerabilities.
- Managed AWS RDS for scalable, secure, and reliable database solutions, reducing manual intervention and cutting database downtime by **20%**.
- Configured and managed AWS WAF to protect web applications against common threats, enhancing security posture and reducing vulnerability exploitation risks.
- Utilized AWS Secrets Manager for secure storage and rotation of credentials, ensuring compliance with security best practices and minimizing exposure risks.
- Implemented Argo CD for automated, GitOps-based continuous delivery, streamlining Kubernetes deployments and improving deployment consistency by **30%**.

- Integrated and managed Azure Active Directory (Azure AD) for secure identity and access management, implementing single sign-on (SSO) and multi-factor authentication (MFA) in one of the project.
- Automated critical system monitoring and deployment workflows using Bash and Python, minimizing manual intervention by **40%** and enhancing deployment consistency.
- Developed AWS Lambda scripts to automate EBS snapshot backups, manage retention policies, remove outdated ECR images, schedule automated log cleanup, and dynamically adjust EC2 instance scaling based on system metrics, enhancing cloud resource efficiency and cost management.
- Led the migration of legacy applications to cloud-native architectures, markedly improving scalability and simplifying operational complexity, resulting in a **15%** reduction in maintenance costs.
- Designed and enforced IAM roles, policies, and RBAC policies on Kubernetes, ensuring secure access control and maintaining compliance across the organization.
- Optimized the deployment and management of 300 Docker containers in production environments, implementing best practices that elevated application stability and reduced deployment errors by **25%**.
- Automated configuration management with Ansible, decreasing manual configuration errors by **35%** and streamlining deployment of microservices and microfrontends in non-Kubernetes environments.
- Designed and implemented a real-time dashboard using AWS QuickSight, leveraging Amazon Redshift as the data warehouse and AWS Glue for ETL processes, enabling seamless access to S3-stored data.
- Deployed and configured the ELK stack to centralize logging and monitoring, accelerating issue detection and resolution by **30%**.
- Created and managed Helm charts for Kubernetes deployments, ensuring consistent and scalable application rollouts across multiple environments.
- Spearheaded release management processes, coordinating multi-environment deployments and maintaining release schedules, improving release accuracy and reducing rollback incidents by **30%**.

Projects

Serverless Web Application | AWS Lambda, API Gateway, DynamoDB, S3

- Engineered a highly scalable serverless web application leveraging AWS Lambda and DynamoDB for seamless and low-latency CRUD operations, optimizing response times via AWS API Gateway.
- Integrated S3 for static website hosting, ensuring smooth frontend-backend interactions and an enhanced user experience with minimal operational overhead.

Data Analytics Platform Deployment | Apache Spark, Elasticsearch, AWS, Terraform

- Designed and deployed a high-performance data analytics platform on AWS using Apache Spark and Elasticsearch, automating data pipelines to achieve a **40%** reduction in processing time.
- Implemented Terraform for Infrastructure as Code (IaC), reducing infrastructure costs by **30%** and integrating real-time monitoring with Prometheus and Grafana for proactive performance optimization.

Microservices-Based Healthcare System | Java, Spring Boot, Docker, Kubernetes, AWS

- Developed a resilient and scalable microservices-based healthcare system using Java and Spring Boot, deploying containerized services via Docker and Kubernetes for enhanced scalability and load balancing.
- Optimized CI/CD pipelines with Jenkins and GitLab, automating deployments on AWS and cutting release time by **50%**, while implementing centralized logging with the ELK stack for improved observability and issue resolution.

Enterprise Cloud Migration and Optimization | AWS, Terraform, Kubernetes, GitLab CI/CD, Ansible

- Migrated enterprise applications from on-premise infrastructure to AWS, improving scalability and cutting operational costs by **40%**.
- Automated infrastructure provisioning and deployment using Terraform, Kubernetes, and GitLab CI/CD, reducing deployment time by **50%** and ensuring high availability.

Achievements & Certifications

AWS Certified Solutions Architect – Associate

Verified by Credly, Badge Link: [AWS Certified Solutions Architect – Associate](#)

HashiCorp Certified: Terraform Associate

Verified by Credly, Badge Link: [HashiCorp Terraform Associate](#)

Certified Kubernetes Administrator (CKA) with Practice Tests By Kodekloud

Cert Link: [Certified Kubernetes Administrator \(CKA\) with Practice Tests](#)

Microsoft Certified: Azure Fundamentals

Cert Link: [Microsoft Certified: Azure Fundamentals](#)